SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I B.Sreesha 192126O034 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sreesha

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 10 marks
Problem Understanding	20%	2	2	2	2	2
Method Selection	20%	2	1	1	2	2
Solution Process	25%	1	2	1	1	1
Accuracy of Calculation	20%	1	1	2	1	1
Interpretation of Results	15%	1	1	1	1	1

Total Marks 35 /40

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1.A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2.A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3.An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4.A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5.A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

1. Problem understanding

Given:

- Original network: `192.168.20.0/24`
- Lab A prefix: `/26`
- Lab B prefix: `/27`
- Lab C prefix: `/28`

The subnets must be allocated without overlapping. Therefore, the next available properly aligned subnet is assigned to each lab.

2. Method selection

The following formulas are used:

- Number of host bits = 32-prefix length
- Total addresses = 2^host bits
- Usable IPv4 hosts = 2^host bits-2

The subtraction of two accounts for the network and broadcast addresses.

• Solution process

Lab A: `/26`

- Subnet mask: `255.255.255.192`

2. Host bits: 32-26=6
3. Total addresses: 2^6=64
4. Usable hosts: 64-2=62
5. Block size: 256-192=64

Therefore, the first subnet covers addresses from `.0` to `.63`. Lab B: `/27`

- Subnet mask: `255.255.255.224`
- Host bits: 32-27=5
- Total addresses: 2^5=32
- Usable hosts: 32-2=30
- Block size: 256-224=32

The next available subnet starts at `.64` and ends at `.95`. Lab C: `/28`

2. Subnet mask: `255.255.255.240`
1. Host bits: 32-28=4

- Total addresses: 2^4=16
- Usable hosts: 16-2=14
- Block size: 256-240=16

The next available subnet starts at `.96` and ends at `.111`.

4. Final results

Details	Lab A	Lab B	Lab C
Prefix	`/26`	`/27`	`/28`
Subnet mask	255.255.255.192	255.255.255.224	255.255.255.240
Network address	192.168.20.0	192.168.20.64	192.168.20.96
First valid host	192.168.20.1	192.168.20.65	192.168.20.97
Last valid host	192.168.20.62	192.168.20.94	192.168.20.110
Broadcast address	192.168.20.63	192.168.20.95	192.168.20.111
Total addresses	64	32	16
Usable hosts	62	30	14

5. Interpretation of results

The three labs receive non-overlapping subnets of different sizes. Lab A supports 62 usable hosts, Lab B supports 30, and Lab C supports 14. The remaining address range from `192.168.20.112` to `192.168.20.255` is available for future allocation.

2: Identifying Subnets for Two Systems

Problem statement: System 1 has IP address `192.168.1.130` with subnet mask `255.255.255.192`. System 2 has IP address `192.168.1.190` with the same subnet mask. Determine the subnet range, network and broadcast addresses, valid host ranges, whether each IP belongs to its subnet, and the number of usable hosts.

•Problem understanding

Given:

- System 1 IP: `192.168.1.130`
- System 2 IP: `192.168.1.190`
- Subnet mask: `255.255.255.192`

The subnet mask corresponds to `/26`. Both systems must be checked against the subnet boundaries determined by this prefix.

2. Method selection

The block size is calculated using:

Block size=256- mask value
=256-192=64

The `/24` address space is divided into four `/26` subnets, with starting addresses `.0`, `.64`, `.128`, and `.192`.

- ## Solution process

The subnet ranges are:

Subnet	Complete address range
1	192.168.1.0–192.168.1.63
2	192.168.1.64–192.168.1.127
3	192.168.1.128–192.168.1.191
4	192.168.1.192–192.168.1.255

Both `192.168.1.130` and `192.168.1.190` fall within the third subnet, which ranges from `.128` to `.191`.

For System 1:

•Network address: `192.168.1.128`
•First valid host: `192.168.1.129`
•Last valid host: `192.168.1.190`
•Broadcast address: `192.168.1.191`
•System IP: `192.168.1.130`

Since `.130` lies between `.129` and `.190`, System 1 has a valid host address. For System 2:

•Network address: `192.168.1.128`
•First valid host: `192.168.1.129`
•Last valid host: `192.168.1.190`
•Broadcast address: `192.168.1.191`
•System IP: `192.168.1.190`

Since `.190` is the last valid host address, System 2 also has a valid host address in the same subnet.

4. Final results

Details	System 1	System 2
IP address	192.168.1.130	192.168.1.190

Subnet mask	**255.255.255.192**	**255.255.255.192**
Prefix	/26	/26
Network address	192.168.1.128	192.168.1.128
Valid host range	192.168.1.129–190	192.168.1.129–190
Broadcast address	192.168.1.191	192.168.1.191
Usable host count	62	62
IP valid in subnet?	Yes	Yes

The number of usable hosts is:

2^(32-26)-2=2^6-2=62

5. Interpretation of results

Both systems belong to the same `/26` subnet, `192.168.1.128/26`. Their IP addresses are valid host addresses and can communicate directly at Layer 2 when they are on the same VLAN or broadcast domain and appropriately configured. No router is required for communication between them under these conditions.

3: Subnet Allocation for Four Departments

Problem statement: An organization has been allocated `172.16.0.0/16`. It must allocate `/25` to HR, `/26` to Sales, `/27` to IT, and `/28` to Admin. Determine the network address, valid host range, broadcast address, usable hosts, and remaining unused address range.

•Problem understanding

Given network: `172.16.0.0/16`. Department requirements:

- HR: `/25`
- Sales: `/26`
- IT: `/27`
- Admin: `/28`

The subnets must be allocated consecutively, respecting each subnet's block size and avoiding address overlap.

•Method selection

For each subnet:

Total addresses=2^(32-p) Usable hosts=2^(32-p)-2

where (p) is the prefix length. The subnet boundaries are determined using the corresponding block sizes.

3.Solution process

HR department — `/25`

- Mask: `255.255.255.128`
- Host bits: 7
- Total addresses: 2^7=128
- Usable hosts: 128-2=126
- Allocation: `172.16.0.0` to `172.16.0.127`

Sales department — `/26`

3. Mask: `255.255.255.192`
4. Host bits: 6
5. Total addresses: 2^6=64
6. Usable hosts: 64-2=62
7. Allocation: `172.16.0.128` to `172.16.0.191`

IT department — `/27`

- Mask: `255.255.255.224`
- Host bits: 5
- Total addresses: 2^5=32
- Usable hosts: 32-2=30
- Allocation: `172.16.0.192` to `172.16.0.223`

Admin department — `/28`

- Mask: `255.255.255.240`
- Host bits: 4
- Total addresses: 2^4=16
- Usable hosts: 16-2=14
- Allocation: `172.16.0.224` to `172.16.0.239`

•Final results

Details	HR	Sales	IT	Admin
Prefix	/25	/26	/27	/28
Network address	172.16.0.0	172.16.0.128	172.16.0.192	172.16.0.224
First valid host	172.16.0.1	172.16.0.129	172.16.0.193	172.16.0.225
Last valid host	172.16.0.126	172.16.0.190	172.16.0.222	172.16.0.238
Broadcast address	172.16.0.127	172.16.0.191	172.16.0.223	172.16.0.239
Total addresses	128	64	32	16
Usable hosts	126	62	30	14

Remaining unused address range:

The total number of allocated addresses is:

128+64+32+16=240

The allocated subnets occupy the first 240 addresses of `172.16.0.0/24`. Therefore, the

remaining unused address range within the original `/16` block is:

`172.16.0.240` through `172.16.255.255`.

This range contains:

2^(16)-240=65,296 addresses

Of these, 65,296 addresses are currently unallocated by the four departmental subnets.

5. Interpretation of results

The organization has allocated 240 addresses to four departments, with 232 usable host addresses across the allocated subnets. The remaining range provides capacity for future departments, additional networks, and organizational expansion. Each subnet has distinct network and broadcast addresses, ensuring that the departmental allocations do not overlap.

4: Subnetting a `/24` Network Using `/27`

Problem statement: A company uses `172.16.10.0/24` and applies a `/27` subnet mask. Calculate the number of subnets and usable hosts per subnet. Identify the subnet containing `172.16.10.78`, its network and broadcast addresses, and determine whether `172.16.10.95` falls within the same subnet.

•Problem understanding

Given:

- Original network: `172.16.10.0/24`
- New prefix: `/27`
- IP address to classify: `172.16.10.78`
- IP address to compare: `172.16.10.95`

The task requires determining how many subnets are created and identifying the correct subnet boundaries.

2. Method selection

The number of borrowed bits is:

27-24=3

The number of subnets is:

2^3= 8subnets

The host bits remaining are:

32-27=5

The total addresses per subnet are:

2^5=32

The usable hosts per subnet are:

32-2=30 hosts

The `/27` subnet mask is `255.255.255.224`, so the block size is 256-224=32.

3. Solution process

The eight subnets are:

Subnet	Network address	Broadcast address
1	172.16.10.0	172.16.10.31
2	172.16.10.32	172.16.10.63
3	172.16.10.64	172.16.10.95
4	172.16.10.96	172.16.10.127
5	172.16.10.128	172.16.10.159
6	172.16.10.160	172.16.10.191
7	172.16.10.192	172.16.10.223
8	172.16.10.224	172.16.10.255

The IP address `172.16.10.78` falls within the third subnet because:

64<=78<=95

Thus, its subnet is `172.16.10.64/27`.

•Network address: `172.16.10.64`
•First valid host: `172.16.10.65`
•Last valid host: `172.16.10.94`
•Broadcast address: `172.16.10.95`

Now consider `172.16.10.95`. It is exactly the broadcast address of the third subnet.

Therefore, it falls within the same subnet's address range, but it is not a valid host IP address. It cannot be assigned to an individual system as a normal host address in this subnet.

4. Final results

Requirement	Answer
Original network	172.16.10.0/24
New subnet prefix	`/27`
Subnet mask	255.255.255.224
Number of subnets	8
Usable hosts per subnet	30
Subnet containing `.78`	172.16.10.64/27
Network address	172.16.10.64
Valid host range	172.16.10.65–172.16.10.94
Broadcast address	172.16.10.95
Is `.95` within the same subnet range?	Yes
Is `.95` a valid host address?	No — it is the broadcast address

5. Interpretation of results

Subnetting the `/24` network into `/27` creates eight equal-sized subnets, each supporting 30 usable hosts. The address `172.16.10.78` is a valid host in the third subnet, while `172.16.10.95` is its broadcast address. This distinction is essential for correct host configuration and avoiding addressing errors.

5: IPv6 Address Compression and Prefix Calculation

Problem statement: Given the IPv6 address `2001:0db8:0000:0000:0000:ff00:0042:8329`, compress it into its shortest form, expand it back to full notation, identify its `/64` network prefix, and calculate the number of possible host addresses.

1. Problem understanding

Given IPv6 address:

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

Prefix length: `/64`.

The requirements are to apply IPv6 zero-compression rules, identify the network prefix, and calculate the number of addresses available in the network.

2. Method selection

IPv6 addresses contain eight groups of hexadecimal digits, called hextets. The following rules are applied:

1. Leading zeros within a hextet can be omitted.
2. One consecutive sequence of all-zero hextets can be replaced by `::`.
3. The double-colon notation can be used only once in a single IPv6 address.

For a `/64` prefix, the first 64 bits identify the network, while the remaining 64 bits are available for interface identifiers.

3. Solution process

Step 1: Write the original address

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

Step 2: Remove leading zeros from each hextet

- `0db8` becomes `db8`

1. `0042` becomes `42`
2. `0000` becomes `0`

The address becomes:

```
2001:db8:0:0:0:ff00:42:8329
```

Step 3: Compress the longest consecutive sequence of zero hextets There are three

consecutive zero hextets. Replace them with `::`.

The shortest form is:

```
2001:db8::ff00:42:8329
```

Step 4: Expand the compressed address

The compressed address contains five explicitly written hextets:

`2001`, `db8`, `ff00`, `42`, and `8329`.

Since IPv6 requires eight hextets, `::` represents three zero hextets. The expanded form is:

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

This matches the original address.

Step 5: Determine the `/64` network prefix
A `/64` prefix covers the first four hextets, or the first 64 bits. Network

prefix:

```
2001:0db8:0000:0000::/64
```

The same prefix in shortened notation is:

```
2001:db8:0:0::/64
```

Step 6: Calculate the number of possible addresses IPv6 addresses contain

128 bits. With a `/64` prefix:

Host bits=128-64=64 Therefore:

Total addresses=2^(64)
=18,446,744,073,709,551,616

A `/64` IPv6 subnet thus contains 2^(64) possible interface-identifier values.

4. Final results

Requirement	Answer
Original IPv6 address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed address	2001:db8::ff00:42:8329
Expanded address	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix length	`/64`
Network prefix	2001:db8:0:0::/64
Network bits	64
Interface identifier bits	64
Possible addresses	18,446,744,073,709,551,616

5. Interpretation of results

IPv6 compression reduces the length of the address without changing its value. The `/64` prefix identifies the network portion, leaving 64 bits for interface identifiers. Unlike conventional IPv4 subnet calculations, IPv6 does not use a broadcast address; therefore, the total number of possible addresses is 2^(64)rather than subtracting two as in the IPv4 examples.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation